

CCNA Day 2: 7/6/26

Ethernet - A collection of network protocols/standards

Bit - 0 or 1

Byte - 8 bits = 1 byte

Speed is measured in bits per second (Kbps, Mbps, Gbps) not bytes per second. Data on a hard drive is measured in bytes. Gigabyte (hence the name)

1 kilobit (Kb) = 1,000 bits

1 megabit (Mb) = 1,000,000 bits (1 Million Bits)

1 gigabit (Gb) = 1,000,000,000 bits (1 Billion Bits)

1 terabit (Tb) = 1,000,000,000,000 bits (1 Trillion Bits)

IEEE 802.3 is ethernet standards

<i>Speed</i>	<i>Common Name</i>	<i>IEEE Standard</i>	<i>Informal Name</i>	<i>Maximum Length</i>
<i>10 Mbps</i>	<i>Ethernet</i>	<i>802.3i</i>	<i>10BASE-T</i>	<i>100 m</i>
<i>100 Mbps</i>	<i>Fast Ethernet</i>	<i>802.3u</i>	<i>100BASE-T</i>	<i>100 m</i>
<i>1 Gbps</i>	<i>Gigabit Ethernet</i>	<i>802.3ab</i>	<i>1000BASE-T</i>	<i>100 m</i>
<i>10 Gbps</i>	<i>10 Gig Ethernet</i>	<i>802.3an</i>	<i>10GBASE-T</i>	<i>100 m</i>

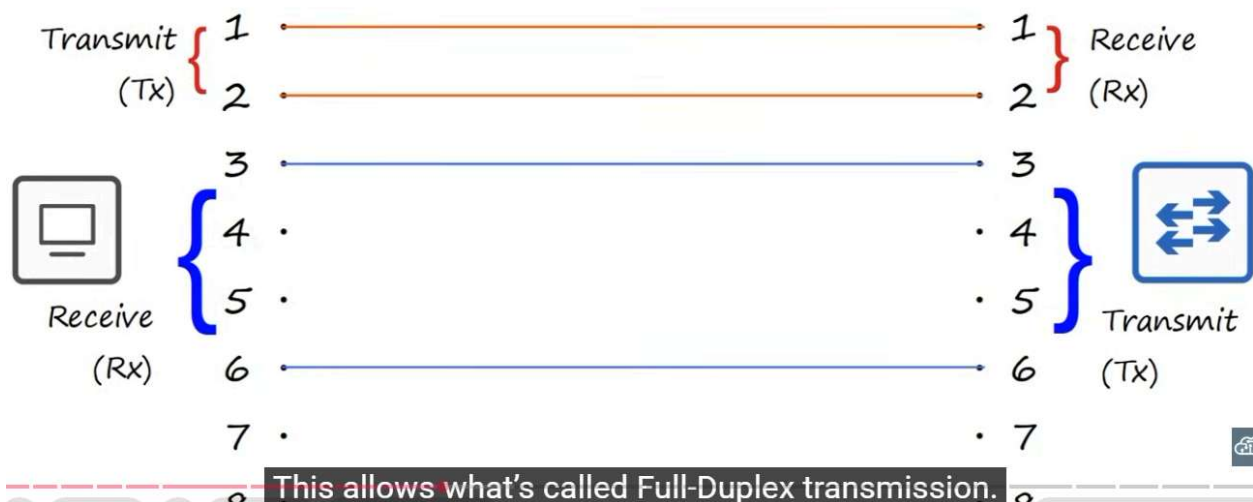
T for informal name means twisted pair. Max length for twisted pair cables are 100m

To put into perspective:



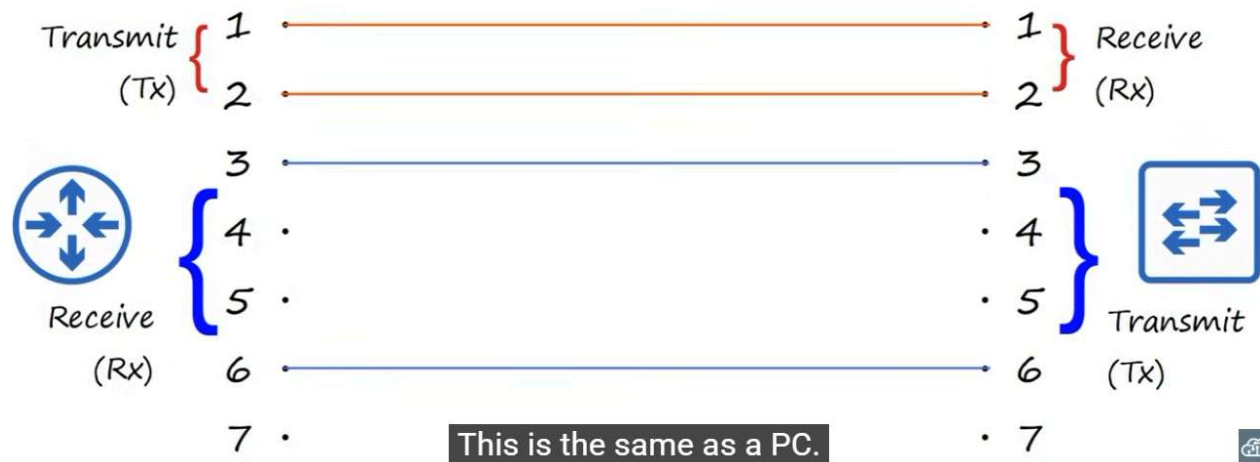
Copper cables: Use UTP cables = Unshielded Twisted Pair (unshielded is having to metallic shield, tinfoil looking thing). Helps with electrical magnetic interference.

Fast ethernet/10Base-T/100Base-T connection (4 cables):



Full Duplex means they can send and receive data at the same time no problem, not having any problems with collision as they use separate wires to transmit and receive data.

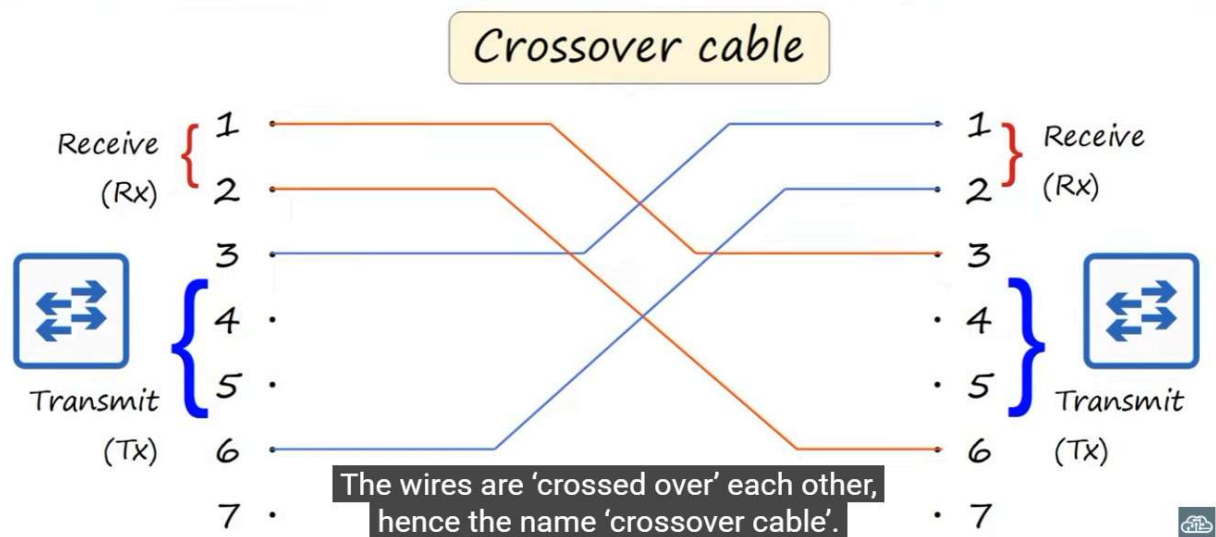
Same setup when it comes to a pc and router:



This is also known as a straight through cable where the same pin connects to its correlating number. ex . 1 to 1, 2 to 2, 3 to 3, and 6 to 6.

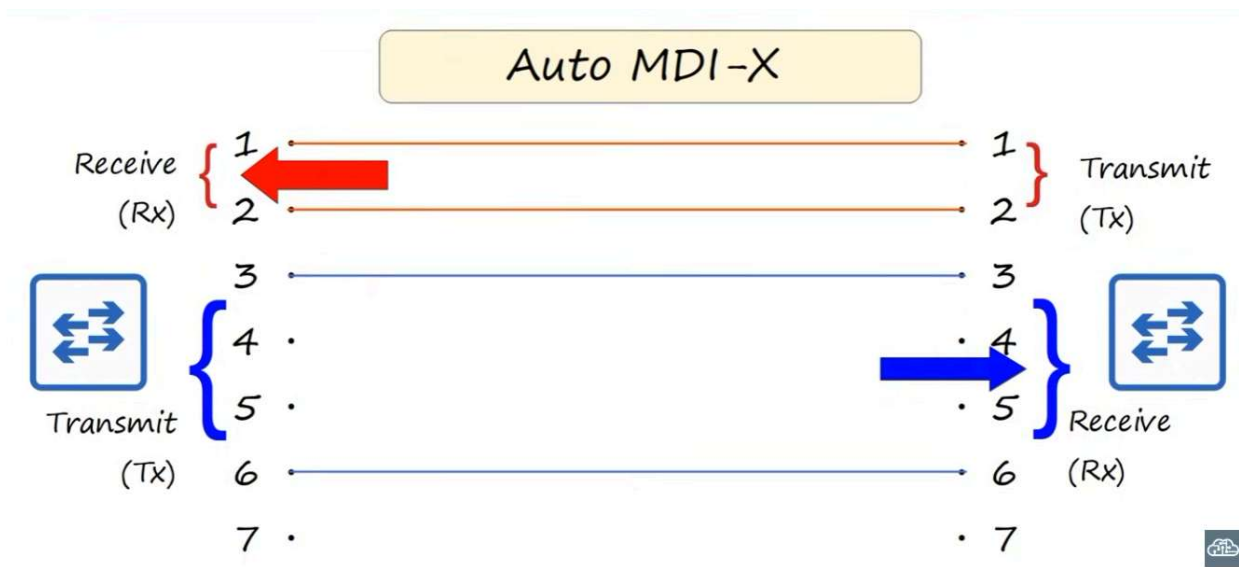
So what if you were to use the same setup with a router to router, pc to pc, or switch to switch?

It will not work since it's not prepared to receive and transmit data. The solution is a cross over cable. In the diagram it shows how the cables "cross over to be able to transmit and receive when both devices transmit and receive on the same interfaces:



Device Type		Transmit (Tx) Pins	Receive (Rx) Pins
Router		1 and 2	3 and 6
Firewall		1 and 2	3 and 6
PC		1 and 2	3 and 6
Switch		3 and 6	1 and 2

In today's world now though, this is basically irrelevant, because modern networking devices have a feature called auto MDI-X. Auto MDIX will actually detect which pins the other device is receiving and transmitting, and then switch to the corresponding input.

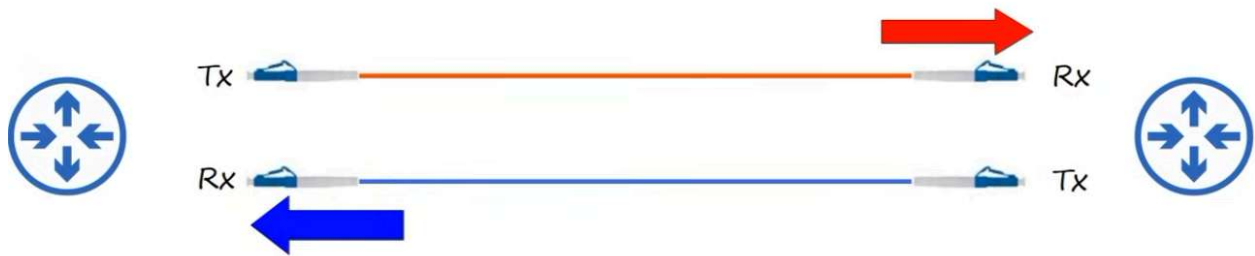


UTP Cables 1000 Base-T and 10G Base-T use the remaining wires, and are bi directional.

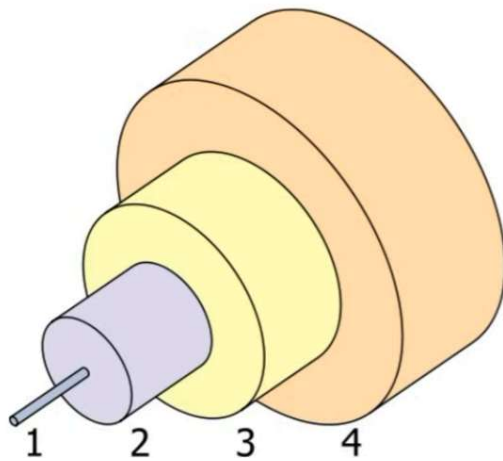
SFP transceiver - Small Form Factor Pluggable:



Usually connect Fiber optic into the SFP:



1. Fiberglass Core
2. Cladding (reflects light)
3. Protective Buffer
4. Outer Jacket of Cable



Single Mode Fiber vs Multi Mode Fiber:

Single-mode is very narrow (8 to 10 μm). Multimode is much larger (50 to 62.5 μm)

Light Source: Single-mode utilizes high-power lasers. Multimode relies on less expensive, broader LEDs or VCSELs

Distance Capabilities: Single-mode can span from 2 km to over 100 km without signal repeaters. Multimode is typically limited to distances under (500-600 meters (up to 2 km for low speeds)).

Single is good for really long distances and high speeds within longer distances. Multimode is good for short distances, cheaper, and high speeds in shorter distances.

Comparison between UTP vs Fiber:

UTP

- Lower cost than fiber-optic.
- Shorter maximum distance than fiber-optic (~100m).
- Can be vulnerable to EMI (Electromagnetic Interference).
- RJ45 ports used with UTP are cheaper than SFP ports.
- Emit (leak) a faint signal outside of the cable, which can be copied (=security risk)

Fiber-Optic

- Higher cost than UTP.
- Longer maximum distance than UTP.
- No vulnerability to EMI.
- SFP ports are more expensive than RJ45 ports (single-mode is more expensive than multimode).
- Does not emit any signal outside of the cable (=no security risk).